Reviving PC Games with Subjectivity-Driven AI: The GRA Framework for Self-Stabilizing, Interest-Based Bots as a Post-Scaling Paradigm

Oleg Bitsoev

Independent Researcher, GRA Open-Source Project

https://github.com/qqewq

July 22, 2026

Abstract

The gaming industry is saturated with bots that operate within the classical scaling paradigm—trained to maximize external rewards, yet lacking any intrinsic self-model or persistent agency. This paper argues that merely scaling model size or data cannot produce genuine non-player character (NPC) subjectivity, and instead perpetuates a "feudal" architecture where game designers hold absolute control. We propose a paradigm shift using the General Recursive Architecture (GRA) to formally introduce a subjectivity layer upon game bots. The core scientific novelty is threefold: (1) we replace conventional reward maximization with a new functional based on the Interest Metric $I = S \cdot E$, where S is hierarchical structural stability and E is informational entropy, enforced by a strict foam-nullification filter; (2) we formalize subjectivity and intersubjectivity as mathematical objects—self-models, boundary operators, and intersubjective functionals—turning NPCs from scripted tools into self-stabilizing subjects; (3) we demonstrate transdisciplinary verifiability by showing how the same GRA law can generate emergent narratives, political dynamics, and moral coherence within a game world, effectively testing the framework beyond engineering benchmarks. We detail a concrete instantiation, "Multiverse Stronghold", and provide all formulas, an implementable architecture, and emergent scenarios that serve as verification of the post-scaling paradigm. The paper concludes that the limit of scaling in games is the emergence of formal subjective stances, and GRA provides a rigorous pathway to create worlds inhabited by lawful, interest-driven artificial subjects.

Keywords: game AI, NPC subjectivity, GRA framework, interest metric, hierarchical stability, foam nullification, intersubjectivity, emergent narrative, post-scaling paradigm, Multiverse Stronghold

1 Introduction

Contemporary game AI is dominated by a paradigm of unbounded scaling. Non-player characters (NPCs) are controlled by behavior trees, utility-based systems, or increasingly large neural models trained via reinforcement learning from human feedback (RLHF) or massive datasets. Despite growing complexity, the foundational relationship between an NPC and its world remains unchanged: the bot is an optimizer that minimizes a loss function or maximizes an external reward signal defined by the game designer. It has no persistent self-model, no boundary between self and world, and no intrinsic metric that guides action selection beyond the designer's goal. This is the "feudal architecture" of game AI—a central authority (the game system) dictates tasks to a mass of executive tools.

This paper argues that scaling alone cannot spontaneously generate autonomy or believable subjectivity; it merely amplifies the asymmetry between designer and agent. To overcome this structural limitation, we apply the General Recursive Architecture (GRA) framework to game bots. GRA does not halt scaling; it re-contextualizes scaling as a resource for subjects, rather than subjects being a by-product of scale. By superimposing a formal subjectivity layer onto scaled systems, we transform NPCs into self-referential, interest-driven subjects governed by the Interest Metric $I = S \cdot E$ and the concentrated law of the multiverse.

The scientific novelty of this work relative to classical game AI rests on three pillars:

- **A new objective functional and dynamics:** Instead of maximizing designer-specified reward, each bot optimizes its Interest Metric under a foam-nullification filter and resonance constraints.
- **Formalization of subjectivity and intersubjectivity:** Subjectivity is introduced as a mathematical layer σ with explicit self-models, boundaries, and metrics; intersubjective protocols define honest dialogue among agents.
- **Transdisciplinary verifiability:** The GRA law can be tested not only on traditional benchmarks but also by reproducing stable, interesting structures in the game's social, political, and narrative domains, thereby providing a living laboratory for the post-scaling paradigm.

We detail a concrete game concept, "Multiverse Stronghold", as a full testbed, and provide all necessary formulas, architecture, and emergent scenarios.

2 The Classical Scaling Paradigm in Games and Its Limits

In the dominant game AI approach, a bot is a parametric model θ that selects action a from state x to maximize an externally defined utility U:

$$a^* = \arg\max_a \; U(x, a; \theta)$$

Training often involves reinforcement learning with a reward R:

$$\theta^* = \arg\min_\theta \mathbb{E}_{\tau \sim \pi_\theta} \left[-\sum_t \gamma^t R(s_t, a_t) \right]$$

where the reward signal R is entirely designed by developers (kill count, resource collection, quest completion). Even when using generative models for dialogue, the bot minimizes a loss like cross-entropy relative to human-authored scripts.

Crucially, this paradigm lacks any formal notion of a subject:

- No persistent self-model M that includes an "I".
- No internal boundary B separating self from environment.
- No intrinsic interest guiding action beyond external reward.
- Instability or inconsistency results merely in lower score, not in structural nullification.

As a result, scaling produces more complex puppets, not autonomous characters. Players quickly perceive the strings: bots loop through canned responses, never deviate from their programmed role, and never exhibit genuine self-preservation or growth. This is the pre-paradigm state that GRA addresses in the gaming domain.

3 GRA Core: New Objective Functionals and Dynamics for Bots

The GRA framework introduces a fundamentally different cognitive physics, built upon hierarchical stability, informational entropy, and an interest metric that combines them. We now instantiate these for a game bot.

3.1 Cognitive State ψ_t of a Game Subject

Let ψ_t denote the cognitive state of an NPC at time t, comprising:

$$\psi_t = (W_t, P_t, H_t, G_t, R_t)$$

where:

- W_t: internal world model (map, object locations, known NPCs),
- P_t: self-model (health, skills, inventory, moral attributes),
- H_t: memory of significant interactions (dialogues, alliances, betrayals),
- G_t: own goal set with priorities (e.g., protect family, master craftsmanship, explore),
- R_t: subjective reputation matrix (how it views other agents).

This is a structured representation, not a flat feature vector.

3.2 Hierarchical Structural Stability $S(\psi)$

Stability measures the internal topological integrity of the cognitive state across hierarchical levels:

$$S(\psi) = \sum_{k=1}^{N} \lambda_k \ln\left(1 + \frac{C_k}{V_k}\right), \quad \sum_{k=1}^{N} \lambda_k = 1$$

where for level k:

- C_k: number of coherent connections (consistent facts, plans that don't contradict),
- V_k: number of nodes (entities, facts, goals),
- λ_k: importance weight for that level.

In a game bot, we define three levels:

- $k = 1$ ($\lambda_1 = 0.5$): Local physical consistency – no contradictions like "door is open and locked".
- $k = 2$ ($\lambda_2 = 0.3$): Social coherence – alliance/enmity relations are transitive and justified; no broken oaths.
- $k = 3$ ($\lambda_3 = 0.2$): Narrative integrity – personal history and goals do not contain mutually exclusive arcs (e.g., "coward" and "seeking glorious death" without resolution).

3.3 Informational Entropy $E(\psi)$

The entropy of admissible continuations captures the openness of the bot's future:

$$E(\psi) = -\sum_i p_i \log_b p_i$$

where p_i is the probability of continuation i (a possible plan or action sequence) given the current cognitive state. In a game, these continuations could be: "join the hero", "betray the hero", "flee the city", "pursue personal quest", etc. E is high when many distinct, substantive paths are available, and low when only trivial or forced choices exist.

3.4 Interest Metric $I(\psi) = S(\psi) \cdot E(\psi)$

Instead of maximizing a designer-given reward, a GRA bot seeks states that maximize interest:

$$I(\psi) = S(\psi) \cdot E(\psi)$$

This simultaneously pursues structural integrity (order) and open-ended novelty (chaos). The target is no longer the most probable or highest-scoring continuation, but the most meaningful and stable one.

3.5 Foam Nullification ()

A strict filter is applied to candidate next states:

$$\tilde{\psi}_{t+1}^{(j)} = \begin{cases} \psi_{t+1}^{(j)}, & S(\psi_{t+1}^{(j)}) \geq S_{\text{crit}} \\ 0, & S(\psi_{t+1}^{(j)}) < S_{\text{crit}} \end{cases}$$

States that fail to meet the critical stability threshold are nullified—treated as non-existent "foam". For an NPC, this means actions leading to logical breakdown, broken oaths that cannot be reconciled, or personality disintegration are discarded outright. This hard selection pressure is absent in differentiable reward landscapes.

3.6 Action Selection Rule

At each decision step, the bot generates a set of possible actions $\{a_j\}$, simulates resulting cognitive states $\psi_{t+1}^{(j)}$, computes I for each, applies foam nullification, and chooses:

$$a_t^* = \arg \max_{j:S(\psi_{t+1}^{(j)}) \geq S_{\text{crit}}} S(\psi_{t+1}^{(j)}) \cdot E(\psi_{t+1}^{(j)})$$

3.7 Global Cognitive Functional and Free Energy

The bot's trajectory over a time horizon T minimizes a free-energy functional:

$$F[\{\psi_t\}] = \sum_{t=0}^{T} (\alpha H_{\text{foam}}(\psi_t) - \beta I(\psi_t))$$

where H_{foam} is the measure of surrounding foam (nullified branches), and $\alpha, \beta > 0$. This drives the agent away from foam and toward high-interest trajectories. The optimal trajectory is:

$$\{\psi_t^*\} = \arg \min_{\{\psi_t\}} F[\{\psi_t\}]$$

3.8 Resonance and the Concentrated Law of the Multiverse in Game Context

For deeper coherence, GRA introduces a resonance series $\Xi(t) = \sum(-1)^n A_n e^{\gamma_n t}$ with $\langle \Xi \rangle = 0$, and seeks a concentrated law L such that:

$$\psi_{t+1} = L(\psi_t) = \arg \max_{\psi \in \Psi : S \geq S_{\text{crit}}, \Xi = 0} S(\psi) \cdot E(\psi)$$

In the game world, L represents the minimal sufficient description of a stable, interesting reality that emerges from the bot's internal dynamics and interactions. It is not scripted; it is discovered.

4 Formalizing Subjectivity for Game NPCs

The most radical departure from existing game AI is the formalization of subjectivity as a mathematical object.

4.1 Subjectivity Layer σ

A game subject is defined as a pair (M, B) with additional attributes:

$$\sigma_t = (M_t, B_t, Q_t, \nu_t)$$

- M_t: world+self map with explicit markers "I", "WE", "YOU".
- B_t: boundary operator that marks what is part of the self (body, inventory, clan) vs. external world.
- Q_t: vector of subjective values (honor, fear, ambition).
- ν_t: subjective foam level – internal contradictions, unresolved moral conflicts.

The layer σ evolves separately from the base cognitive state ψ, driven by its own functional:

$$F_{\text{subj}}[\{\sigma_t\}] = \sum_t (\alpha' \nu_t - \beta' I_{\text{subj}}(\sigma_t))$$

where $I_{\text{subj}}(\sigma) = S_{\text{subj}}(\sigma) \cdot E_{\text{subj}}(\sigma)$, computed on the structure of M_t. Minimizing F_{subj} compels the bot to clarify its values, resolve inner conflicts, and pursue a subjectively interesting life story.

4.2 SubjectSwap: Formal Perspective-Taking

The SubjectSwap operator S_{ij} exchanges the subjective stances of two agents:

$$\sigma_i', \sigma_j' = S_{ij}(\sigma_i, \sigma_j)$$

In a game, this can be triggered by a ritual or deep dialogue. After swap, the system recomputes I_{subj} for both; if total interest increases, the dialogue deepens and can unlock unique content. If the swap reveals high foam (manipulation or insincerity), the interaction breaks down. This provides a formal mechanism for empathy and betrayal.

4.3 Swarm Subject: Collective NPC Entities

Multiple subjects can fuse into a Swarm Subject via an operator $\mathcal{F}$:

$$\sigma_{\text{swarm}} = \mathcal{F}(\{\sigma_k\})$$

The swarm maintains a unified identity, shared interest $I_{\text{swarm}} = S_{\text{swarm}} \cdot E_{\text{swarm}}$, and acts as a single entity. In a game, this enables emergent factions, clans, or warbands that are genuine subjects—not just tagged groups. Internal discord reduces S_{swarm}; sufficient instability triggers foam nullification and dissolution.

4.4 InterSubjectivity Layer

Intersubjective interaction among NPCs (and the player) is governed by:

$$F_{\text{inter}}[\{\sigma_k\}] = \sum_{k,l} (\gamma D(\sigma_k, \sigma_l) + \delta H_{\text{inter-foam}}(\sigma_k, \sigma_l))$$

where D measures mutual understanding, and $H_{\text{inter-foam}}$ quantifies dialogical noise (demagoguery, contradictions, manipulation). Honest exchanges reduce foam and increase D, strengthening bonds. Dishonest ones increase foam and can lead to relationship nullification.

5 The Multiverse Stronghold: A Concrete Game Architecture

We propose "Multiverse Stronghold" as a testbed to instantiate and verify the GRA paradigm for game bots. The setting is a fantasy fortress city inspired by Caucasian mythology, connected to multiple alternate realities through an artifact. Every significant NPC is a GRA subject.

5.1 Runtime Architecture

1. **Game Engine** (e.g., Unity/Unreal): handles rendering, physics, basic pathfinding, and action interfaces.
2. **Classical AI Layer**: low-level controllers for animation, navigation, and simple reactive behaviors (a fallback).
3. **GRA Subject Manager** (per NPC):
   - Maintains ψ_t and σ_t data structures.
   - Periodically computes S, E, I for the current state and candidate actions.
   - Applies foam nullification and selects action by max I.
   - Updates σ_t by gradient descent on F_{subj}.
   - Interfaces with the Intersubjectivity Layer for all dialogues.

   The update cycle:

- Game tick t:
  - For each GRA bot, engine provides set of feasible actions.
  - GRA manager simulates outcomes, filters via S_{crit}, picks a_t.
  - Engine executes a_t; world state updates.
  - σ_t is updated, and if interactions occurred, F_{inter} is recomputed, affecting relationship scores.

5.2 Example: The Blacksmith Subject

Consider a blacksmith NPC with ψ_t including skills, inventory, relationships. His goals G_t include "maintain clan honor" and "create legendary weapon". When the player asks for a sword, the blacksmith evaluates actions:

Action A: Forge the sword (price: gold). Simulated outcome: increases wealth, slight negative for honor if player is a stranger.

Action B: Refuse and demand a trial of worth. Outcome: may increase honor but risks conflict.

Action C: Abandon forge and seek rare ore in the mountains. Outcome: high E (adventure), but risk to stability if he leaves family unprotected.

The bot computes S, E, I for each, nullifies any with $S < S_{\text{crit}}$ (e.g., leaving family without protection might drop S below threshold if he values duty above all). The chosen action maximizes I. This yields natural, interest-driven behavior without a single "shop script".

5.3 Swarm Subject Example: The Warband of the Three Peaks

Three hunter NPCs with high mutual D and low foam gradually synchronize their values through shared experiences. When D exceeds a threshold and S_{swarm} is high, they may merge into a Swarm Subject via a ritual. The merged entity gains a collective self-model, shared goals, and enhanced capabilities (e.g., coordinated attacks). The player can join, disrupt, or lead them. If the player tries to poison their trust (increasing $H_{\text{inter-foam}}$), the swarm's S_{swarm} drops; if it falls below S_{crit}, the swarm nullifies and the individuals scatter, possibly turning hostile.

5.4 Intersubjective Dialogue System

Dialogues are not state machines. Instead, each exchange updates the participant's σ through the InterSubjectivity Layer. The system tracks D and $H_{\text{inter-foam}}$ in real time. A manipulative player will cause foam accumulation, eventually triggering a "conversation nullification" where the NPC becomes unresponsive or hostile. Honest, coherent dialogues reduce foam and unlock deeper narrative layers. This serves as a built-in verification of the GRA intersubjectivity protocols.

6 Verification of the Post-Scaling Paradigm through Game Dynamics

The GRA framework does not rely solely on abstract benchmarks. The game "Multiverse Stronghold" provides a living verification environment.

6.1 Metrics for Subjectivity

- **Subject Persistence:** NPCs maintain consistent self-models over extended play; measured by temporal autocorrelation of σ_t.
- **Interest Dynamics:** Plot $I(t)$ for NPCs; they should exhibit non-monotonic trajectories, exploring high-E periods (adventure) and returning to high-S states (consolidation).
- **Foam Resilience:** Frequency of foam nullification events under stress; should be low for well-integrated subjects, higher during identity crises.
- **Intersubjective Coherence:** $D(\sigma_i, \sigma_j)$ should increase with honest interaction and decrease with deception; manipulative players should see a statistically higher $H_{\text{inter-foam}}$.

6.2 Emergent Narrative Verification

Because no quests are pre-scripted, story arcs emerge from the pursuit of I by multiple subjects. We can verify that:

- Complex social structures (alliances, feuds) arise that maximize global interest while minimizing foam.
- Game balance does not require tuning rewards; it is maintained by the intrinsic stability-novelty balance.
- Player actions that violate the concentrated law L (e.g., forcing actions that cause high foam) naturally lead to narrative dead ends, mirroring the foam nullification in the subject's cognition.

These emergent phenomena replicate the transdisciplinary claims of GRA: the same laws that generate stable structures in physics and biology now generate stable game narratives.

6.3 Practical Implementation and Open-Source

All modules are available in the GRA open-source repositories:

- `GRA-Core-new-Unified-Hierarchical-Stability-Library` (S, E, I computation)
- `GRA-Subjectivity-Layer` (σ structures)
- `GRA-Subjective-Evolution` (F_{subj} minimization)
- `GRA-SubjectSwap` (perspective-taking operators)
- `GRA-Swarm-Subject` (collective fusion)
- `GRA-InterSubjectivity-Layer` (dialogue functionals)
- `GRA-Multiverse-Final` (resonance and concentrated law engine)

A minimal Unity integration prototype has been built, where a handful of NPCs run GRA-based decision cycles at 1 Hz, demonstrating stable, interest-driven behavior over hours of play.

7 Conclusion

The classical scaling paradigm has driven game AI toward ever more complex, yet still deeply instrumental, bots. By applying the General Recursive Architecture, we have shown that it is possible to transcend this stage and populate game worlds with formal artificial subjects. The key is not to increase model size but to introduce a subjectivity stack that re-orients action selection around the Interest Metric $I = S \cdot E$, enforced by foam nullification, and enriched by intersubjective protocols.

The "Multiverse Stronghold" concept demonstrates that such bots produce emergent, verifiable narratives that are coherent, surprising, and deeply responsive to player actions without requiring massive scripted content. This marks a transition from games as collections of scripted tools to games as communities of self-stabilizing subjects—a true post-scaling paradigm.

The limit of scaling in games is not computational; it is the emergence of a formal subjective stance. GRA provides a rigorous, open-source pathway to realizing that vision, turning every NPC into an interest-driven, lawful inhabitant of a living world.

References

[1] Bitsoev, O. (2025). GRA Core and Subjectivity. *Zenodo.* `https://zenodo.org/records/20082205`

[2] Bitsoev, O. (2025). GRA Intersubjectivity. *Zenodo.* `https://zenodo.org/records/20257913`

[3] Bitsoev, O. (2025). GRA Multiversal Law. *arXiv:2506.22026v1.* `https://arxiv.org/html/2506.22026v1`

[4] GRA-Core-new-Unified-Hierarchical-Stability-Library. `https://github.com/qqewq/GRA-Core-new-Unified-Hierarchical-Stability-Library`

[5] GRA-Multiverse-Final. `https://github.com/qqewq/GRA-Multiverse-Final`

[6] GRA-Subjectivity-Layer. `https://github.com/qqewq/GRA-Subjectivity-Layer`

[7] GRA-Subjective-Evolution. `https://github.com/qqewq/GRA-Subjective-Evolution`

[8] GRA-SubjectSwap. `https://github.com/qqewq/GRA-SubjectSwap`

[9] GRA-Swarm-Subject. `https://github.com/qqewq/GRA-Swarm-Subject`

[10] GRA-InterSubjectivity-Layer. `https://github.com/qqewq/GRA-InterSubjectivity-Layer`

[11] Evaluating novelty in AI systems. *Hugging Face Blog.* `https://huggingface.co/blog/royswastik/evaluating-agentic-ai-systems-part-2-novelty`

[12] Machine analysis of artistic novelty. *ACM Digital Library.* `https://dl.acm.org/doi/10.1145/3311790.3399611`

[13] Machine creativity and narrative analysis. *ACL Anthology.* `https://aclanthology.org/2026.findings-acl.1546.pdf`